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INVENTIVE DISCLOSURE - CONFIDENTIAL

1.	Title of the Invention
	Context-Aware Smart Kitchen Safety System with Dual-Microcontroller Architecture, IoT Dashboard, and Real-Time Alerting
2.	What is the Technical Field of Invention?
	This invention relates to the field of Internet of Things (IoT)-based safety systems, embedded systems, and smart home automation, specifically focusing on kitchen safety monitoring and automated hazard prevention. Problem: Existing IoT kitchen safety systems depend heavily on cloud connectivity and single-controller architectures, making them unreliable during network failures. Solution: The invention introduces a dual-microcontroller architecture that separates: ● Safety-critical processing (edge layer) ● Network and user interface operations This ensures fail-safe operation even without internet connectivity.
3.	Search Terms
	● Smart kitchen safety system ● IoT gas leak detection ● Dual microcontroller IoT architecture ● Edge computing safety systems ● Sensor fusion kitchen monitoring ● Automatic gas shutoff system

	● Embedded safety controller
4.	Background of the Invention (Present state of Art)
	a. What are the present technologies that exist in the field of your invention Single-chip IoT systems (e.g., ESP8266, ESP32) Gas leak detectors with alarms Smoke detectors Smart home automation apps
	b. What are the limitations of the same? Dependence on Wi-Fi/cloud No preemptive safety logic No separation of safety and communication layers Delayed response in hazardous situations
5.	Prior Art
	Existing systems: ● Gas detectors trigger alarms only after detection ● Smart IoT systems rely on internet availability ● Fire alarms act after ignition Limitation: No system provides local, autonomous decision-making with hardware-level override capability.
6.	Has the work filled a major gap in the prior art? If yes, a brief description of this gap.
	Yes. There is a major gap in: ● Fail-safe offline safety operation ● Sensor fusion for predictive hazard detection ● Hardware-level authority overriding remote commands
7.	What are the components of the Invention?
	The proposed invention consists of a modular, dual-microcontroller architecture divided into hardware subsystems, communication interfaces, and software-controlled functional blocks.

A. External Components (User-Visible / Physical Layer)

These are the components directly interacting with the environment:

1. **Gas Sensor (Analog)**
 - Detects gas concentration levels in the kitchen environment.
 2. **Temperature and Humidity Sensor (DHT11)**
 - Measures ambient temperature and humidity.
 3. **Flame Sensor**
 - Detects presence of open flame.
 4. **PIR Motion Sensor**
 - Detects human movement to determine occupancy.
 5. **Servo Motor (Gas Valve Control)**
 - Mechanically controls gas supply (open/close).
 6. **Relay Module (Exhaust Fan Control)**
 - Activates ventilation system when required.
 7. **Buzzer**
 - Provides audible alerts during hazardous conditions.
 8. **Indicator LEDs**
 - Alert LED (blinking during hazard)
 - Motion LED (indicates occupancy)
 9. **Push Button**
 - Allows manual override for gas valve control.
-

B. Internal Components (Processing & Control Layer)

1. Primary Safety Controller – Raspberry Pi Pico

- Acts as the **central decision-making unit**
- Executes real-time safety logic
- Directly interfaces with sensors and actuators
- Operates independently of network connectivity

2. Secondary Network Controller – ESP8266

- Handles **Wi-Fi communication and IoT functionalities**
 - Hosts WebSocket-based dashboard
 - Sends alerts via Telegram API
 - Communicates with Pico via UART
-

C. Communication Subsystem

- **UART Serial Communication**
 - Bi-directional communication between Pico and ESP8266
 - Transfers:
 - Sensor data (JSON format)
 - User commands (control instructions)
-

D. Functional Subsystems

1. Sensor Acquisition Module

- Periodically reads:
 - Gas levels
 - Temperature
 - Flame status

	○ Motion status 2. Safety Decision Engine ● Implements rule-based logic: ○ Gas threshold detection ○ Flame detection ○ Temperature monitoring ○ Unattended cooking detection 3. Actuation Control Module ● Controls: ○ Gas valve (servo motor) ○ Exhaust fan (relay) ○ Buzzer alerts ○ LED indicators 4. Manual Override Module ● Processes push-button input ● Allows user to toggle valve (unless unsafe condition exists) 5. Network Interface Module (ESP8266) ● Provides: ○ Real-time dashboard ○ Remote control interface ○ Telegram alert notifications 6. Alert Management System ● Generates alerts for: ○ Gas leak (GAS_HIGH) ○ Flame detection (FLAME) ○ Unattended cooking (UNATTENDED) E. Process Flow (System Operation) 1. Sensors continuously collect environmental data 2. Data is processed by the Pico safety controller 3. Safety rules are evaluated 4. If hazard detected: ○ Immediate action is taken (valve close, alarm, etc.) 5. System state is transmitted to ESP8266 6. ESP8266: ○ Updates dashboard ○ Sends alerts to user 7. User commands (if any) are sent back to Pico 8. Pico validates commands: ○ Executes only if safe ○ Rejects unsafe commands
8.	How does the Invention work? <i>(This portion of your invention disclosure deals with the working of your invention, the processes involved and the interactive actions taking place between the various components of your invention to give the desired output.)</i>
	The system operates using two independent processing layers:

1. Safety Layer (Raspberry Pi Pico)

- Continuously reads sensor data every ~1 second
- Applies safety rules:
 - Gas threshold detection
 - Flame detection
 - Temperature monitoring
 - Motion detection
- Executes actions:
 - Closes gas valve (servo)
 - Activates buzzer
 - Controls exhaust fan
- Works **independently of internet**

2. Network Layer (ESP8266)

- Connects to Wi-Fi
- Hosts WebSocket dashboard
- Sends alerts via Telegram API
- Displays real-time sensor data

3. Key Interaction Mechanism

- ESP8266 communicates with Pico via UART
- Pico sends JSON data periodically
- ESP sends user commands to Pico

4. Safety Override Logic (Critical Innovation)

- If unsafe condition is detected:
 - Pico **ignores all external commands**
 - Enforces safety actions locally

5. Sensor Fusion Logic

Example:

- Temperature high + No motion → system deduces unattended cooking
- Automatically shuts gas valve

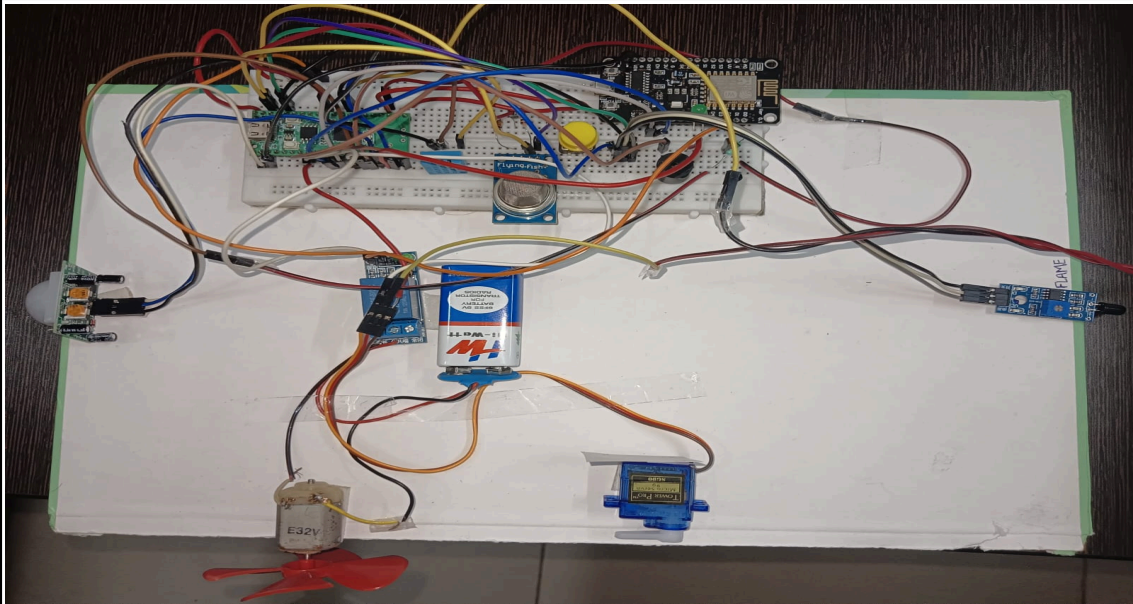
9.

Novelty of the Invention/Inventive Step of the Invention

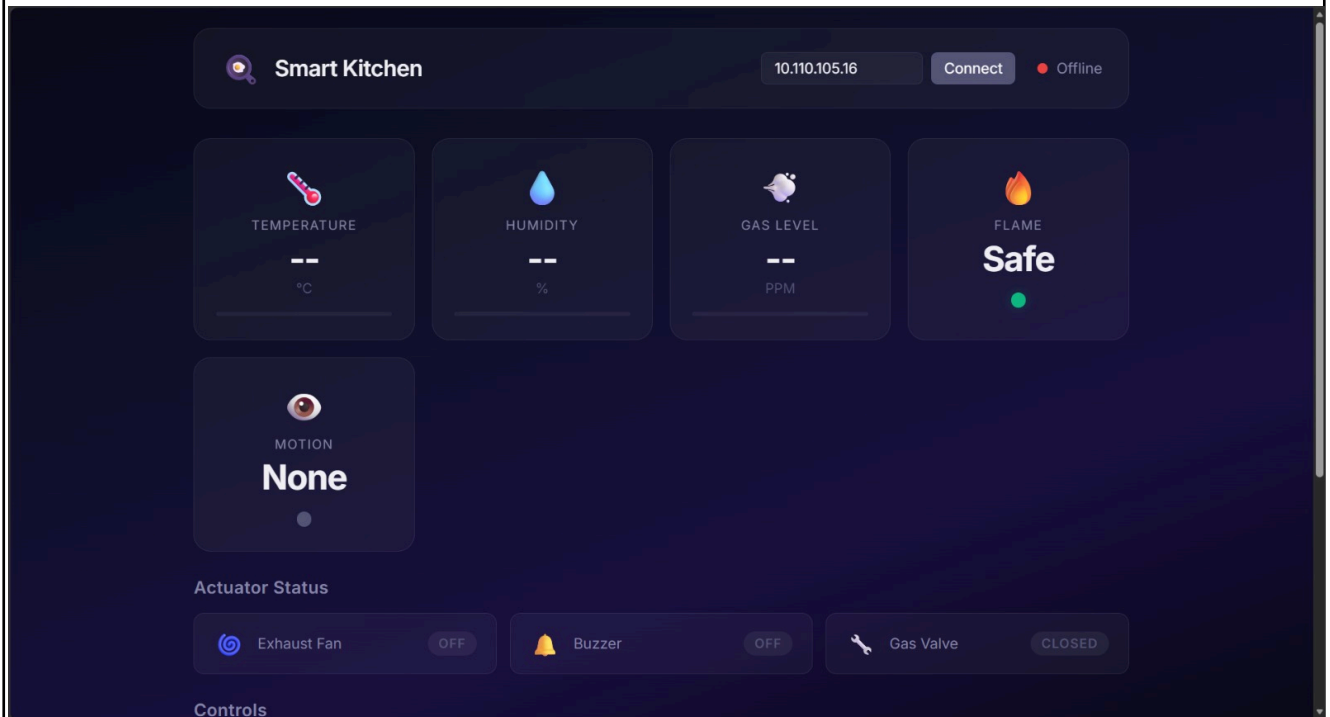
The invention introduces:

1. **Dual-Microcontroller Architecture**
 - Separation of safety and networking
2. **Hardware-Level Override Authority**
 - Safety controller overrides remote commands
3. **Sensor Fusion-Based Prediction**
 - Detects unsafe conditions before accidents occur

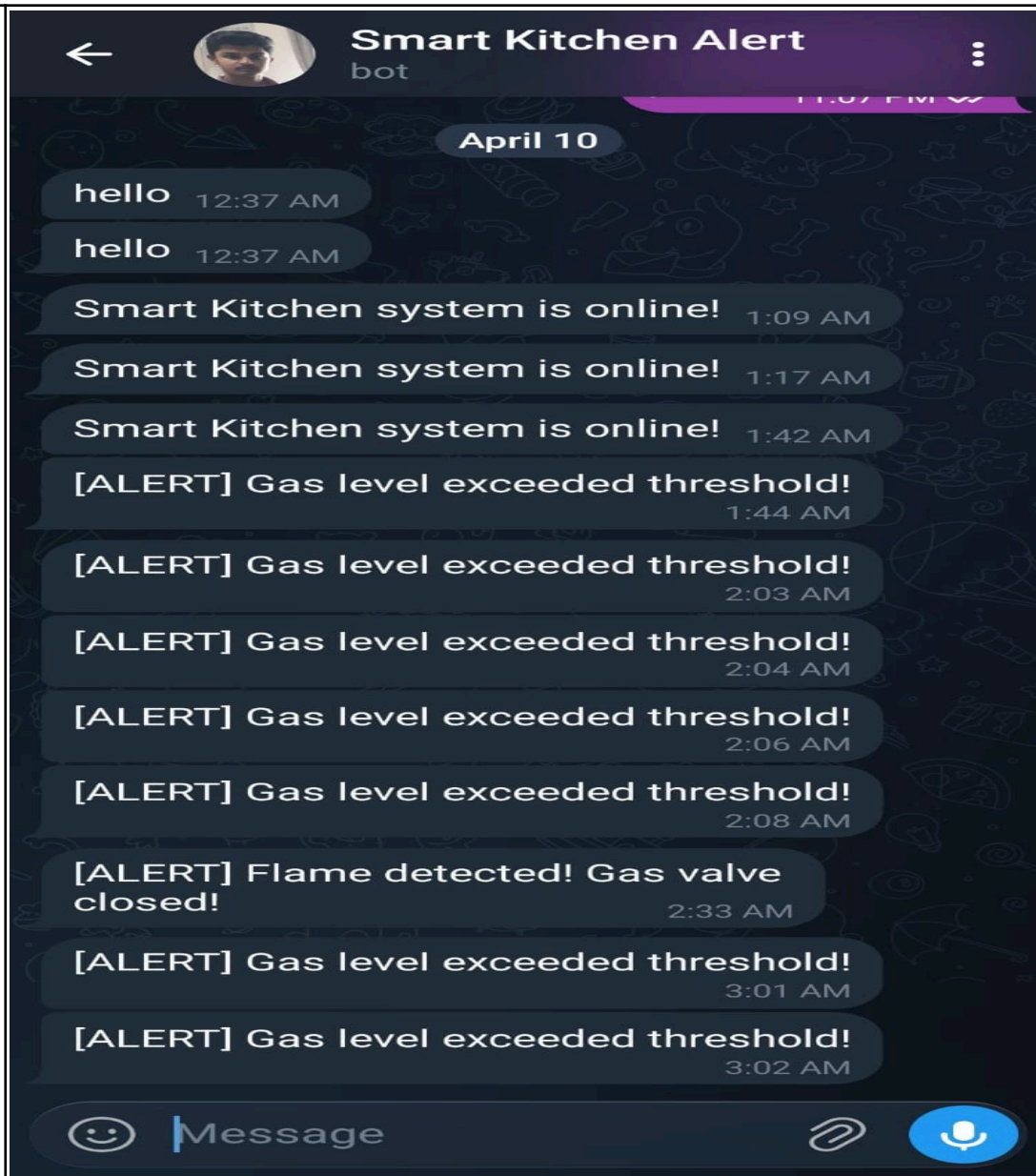
	4. Fail-Safe Offline Operation ○ Works even without Wi-Fi
10.	What are the advantages and improvements over the existing methods/devices?
	The proposed invention provides the following key improvements over existing gas detection and IoT-based safety systems: ● Fail-safe operation: Unlike single-controller systems, the dual-microcontroller design ensures safety functions continue even if network/Wi-Fi fails. ● Hardware-level safety override: The safety controller (Pico) has final authority and can reject unsafe remote commands, preventing dangerous actions. ● Predictive safety (sensor fusion): Combines gas, temperature, and motion data to detect conditions like unattended cooking before accidents occur, unlike traditional reactive systems. ● Automatic hazard response: Not only detects danger but also takes immediate action (gas valve shutoff, fan activation, alarm), whereas many systems only provide alerts. ● Reduced false alarms: Multi-condition logic improves accuracy compared to single-sensor threshold-based systems. ● Separation of safety and network layers: Improves system stability and reliability by isolating critical safety operations from network processes. ● Faster response (edge processing): Local decision-making eliminates cloud delay and ensures real-time action. ● Multi-channel alerting: Provides real-time monitoring via dashboard and Telegram alerts, improving user awareness. ● Scalable and modular design: Easily extendable with additional sensors or advanced features.
11.	Do you have in your mind as a researcher/inventor some ideas as to what are the various other/alternative forms in which your Invention can be depicted or utilized, in addition to one basic embodiment that has been described above? <i>(This will lay the foundation, for making your patent as broad as possible. These are called preferred embodiments or variants.)</i>
	Yes, the invention can be implemented in several alternative forms and extended embodiments: ● The dual-controller architecture can be implemented using other microcontrollers such as ESP32 or industrial-grade controllers. ● The system can be adapted for industrial kitchens, gas storage units, or laboratories with higher-capacity sensors and actuators. ● Additional sensors (smoke, CO, LPG-specific sensors, camera modules) can be integrated for enhanced safety monitoring. ● AI/ML algorithms can be incorporated for advanced anomaly detection and predictive maintenance. ● The alert system can be extended to include mobile applications, SMS, email, or voice assistants. ● The system can be redesigned as a fully integrated single-chip solution with logically separated safety and network tasks. ● Renewable power sources (battery/solar) can be added for uninterrupted operation during power failures.
12.	Please provide final CAD images/line diagrams of the final prototype which illustrates all the essential elements of the invention. <i>(Please prepare images in two sets, one completely labeled to details and the second one without any description, legends/labeling or numbering. Include relevant Flow charts.)</i>
	PROTOTYPE IMAGE:



DASHBOARD SNAPSHOT:



TELEGRAM BOT(ALERTS):



13. Brief description from the researcher/inventor in his own words explaining the images/ drawings.

The diagrams illustrate the overall architecture and working of the proposed smart kitchen safety system. The system is divided into two main sections: the **safety control unit (Raspberry Pi Pico)** and the **network communication unit (ESP8266)**.

The drawings show how various sensors such as gas, temperature, flame, and motion are connected to the Pico, which continuously monitors environmental conditions and controls actuators like the gas valve (servo), exhaust fan (relay), buzzer, and indicator LEDs.

The ESP8266 is connected to the Pico through a UART interface and is responsible for Wi-Fi connectivity, real-time dashboard updates, and sending alerts to the user. The data flow between components and the decision-making process of the system are represented in the diagrams.

Overall, the drawings explain how the system detects hazardous conditions and automatically takes safety actions while also providing remote monitoring and alerting features.

14.	Please give complete description including the working examples/ working methodology of the invention.
	Complete Description and Working Methodology The proposed invention is a dual-microcontroller based smart kitchen safety system designed to detect hazards and take immediate preventive actions. Working Methodology: The system operates in a continuous loop where the Raspberry Pi Pico (safety controller) reads data from multiple sensors including gas, temperature, flame, and motion. Based on predefined threshold values and logical conditions, it evaluates the safety status of the environment. • If gas concentration exceeds the threshold, the system immediately closes the gas valve, activates the exhaust fan, and triggers a buzzer alert. • If a flame is detected, the gas supply is cut off and an alert is generated. • If high temperature is detected, the exhaust fan is activated for cooling. • If high temperature is present without human motion for a specific duration, the system identifies it as unattended cooking and automatically shuts off the gas supply. All control actions are executed locally by the Pico, ensuring real-time response and independence from network conditions. Simultaneously, the ESP8266 (network controller) receives processed data from the Pico via UART communication. It updates a real-time dashboard and sends alert notifications to the user through Telegram. The system also allows manual control (such as opening/closing the gas valve), but any unsafe command is overridden by the safety controller to ensure protection. Working Example: For instance, if a user leaves the stove ON and exits the kitchen, the system detects high temperature and absence of motion for a predefined time. It then automatically turns OFF the gas valve and sends an alert message to the user, preventing a potential fire hazard.
15.	Please provide details of experiments conducted /third party validation data if any.
	The prototype was tested under controlled conditions using different scenarios: • Gas leakage test: Gas sensor was exposed to varying concentrations; system successfully detected threshold breach and automatically closed the gas valve and activated alerts. • Flame detection test: Flame presence was introduced; system immediately shut off gas supply and triggered alarm. • Temperature test: Elevated temperature conditions were simulated; exhaust fan was activated as expected. • Unattended cooking test: High temperature with no motion for a defined time resulted in automatic gas shutoff. • System reliability test: Communication between controllers and alert delivery (Telegram) was verified. No third-party validation has been conducted yet other than my faculty ; testing has been performed at prototype level.
16.	Please provide details of any public disclosure done/ any publication made.

	No public disclosure or formal publication has been made for this invention. The details are currently confidential and have only been used for academic/project purposes.																										
17.	What is the stage/level of development of the Invention?																										
	a. At a basic conceptualization stage? The invention was initially designed as a concept focusing on improving reliability of existing IoT safety systems through a dual-controller architecture and predictive safety logic.																										
	b. Completed and results validated? A functional prototype has been developed and tested successfully, demonstrating real-time hazard detection, automatic safety actions, and reliable operation.																										
18.	What are the aspects of your disclosure that you want to claim/monopolize? (Proposed Claims)																										
	A dual-microcontroller based safety system separating safety-critical processing and network communication. A fail-safe control mechanism where the safety controller operates independently of internet connectivity. A hardware-level override system that rejects unsafe user commands and enforces protective actions. A sensor fusion-based hazard detection method combining gas, temperature, and motion data to predict unsafe conditions. An automatic gas shutoff and ventilation control system triggered by detected hazards. A real-time monitoring and alert system integrated with IoT dashboard and messaging services.																										
19.	Have you conducted novelty/inventiveness search for your invention? If yes, what are the databases/references used by you? What are the search results?																										
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20.	Do you feel that a person of “average” skill (not-extraordinary skill) in your area of technology would have arrived at your invention with existing knowledge in public domain? If no, what could be the reasons for the same?			
	No. A person of average skill would not easily arrive at this invention because it combines dual-microcontroller architecture, fail-safe operation, and sensor fusion in a non-obvious way. Existing systems are mostly single-controller and reactive, whereas this invention introduces independent safety control with hardware-level override and predictive hazard detection, which is not common in current technologies.			
21.	Kindly provide broad workable ranges for all the parameters involved in your invention.			
	The invention operates within the following broad parameter ranges: ● Gas Concentration Threshold: 1000 – 10000 ppm ● Temperature Range: 0°C – 80°C ● Temperature Threshold (Safety Trigger): 40°C – 70°C ● Humidity Range: 20% – 90% RH ● Unattended Time Duration: 5 – 15 minutes ● Sensor Sampling Interval: 0.5 – 2 seconds ● Alert Cooldown Time: 20 – 60 seconds ● Servo Angle (Valve Control): 0° (closed) to 90° (open) ● Operating Voltage: 3.3V – 5V ● Communication Baud Rate (UART): 9600 bps			
22.	Commercialization Data			
	a. Kindly give the names and complete addresses of five different companies which might be interested in the commercial manufacture of this technology. 1. Honeywell Automation India Ltd., Unitech Trade Centre, Sector 43, Gurugram, Haryana, India 2. Bosch Limited, Hosur Road, Adugodi, Bengaluru, Karnataka, India 3. Siemens Limited India, Birla Aurora, Dr. Annie Besant Road, Worli, Mumbai, Maharashtra, India 4. Schneider Electric India Pvt. Ltd., DLF Building No. 10, Tower C, Gurugram, Haryana, India 5. Havells India Ltd., QRG Towers, Sector 126, Noida, Uttar Pradesh, India			
	b. Kindly also enclose a short marketing profile of the technology highlighting the advantages vis-à-vis the prior-art technologies in this area The proposed technology is a smart, fail-safe kitchen safety system that provides real-time hazard detection and automatic preventive action. Unlike existing systems that only generate alerts, this invention actively prevents accidents by shutting off gas supply and controlling ventilation. Its dual-microcontroller architecture ensures reliability even without internet connectivity, making it highly suitable for real-world deployment. The system is low-cost, scalable, and easy to integrate into existing kitchens, giving it a strong competitive advantage over traditional gas detectors			

	and IoT-based safety devices.																		
23.	Provide available data or information on market potential/size, adoption trends, industry analysis, and projected growth.																		
	The market for smart safety and IoT-based home systems is rapidly growing due to increasing awareness of fire and gas hazards, urbanization, and adoption of smart home technologies. The proposed invention aligns with these trends by offering a low-cost, intelligent, and fail-safe solution, making it highly suitable for future market demand. <table border="1"> <thead> <tr> <th>Year</th><th>Estimated Market Size</th><th>Growth Drivers</th><th>Notable Existing Devices</th></tr> </thead> <tbody> <tr> <td>2023</td><td>~\$80 Billion (Global Smart Home Market)</td><td>Increasing IoT adoption, safety awareness, urbanization</td><td>Nest Protect, Xiaomi Gas Detector</td></tr> <tr> <td>2025</td><td>~\$110 Billion (Projected)</td><td>Growth in smart kitchens, rising accident prevention demand</td><td>Honeywell Gas Leak Detector, Kidde Smoke Alarm</td></tr> <tr> <td>2030</td><td>~\$170 Billion (Projected)</td><td>AI integration, smart automation, government safety regulations</td><td>Bosch Smart Home Devices, Tuya IoT Systems</td></tr> </tbody> </table>			Year	Estimated Market Size	Growth Drivers	Notable Existing Devices	2023	~\$80 Billion (Global Smart Home Market)	Increasing IoT adoption, safety awareness, urbanization	Nest Protect, Xiaomi Gas Detector	2025	~\$110 Billion (Projected)	Growth in smart kitchens, rising accident prevention demand	Honeywell Gas Leak Detector, Kidde Smoke Alarm	2030	~\$170 Billion (Projected)	AI integration, smart automation, government safety regulations	Bosch Smart Home Devices, Tuya IoT Systems
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24.	Please provide the names of the inventors involved in the Invention																		
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25.	User Information																		
	a. Who all are the potential users of the technology? - Household users (homes, apartments) - Restaurants and commercial kitchens - Hostels and shared accommodations - Gas storage and small industrial units																		
	b. Age group of the user? Applicable to all age groups (18–65+), especially useful for families, elderly people, and busy professionals																		
	c. Expected benefits to the user? - Prevention of gas leaks and fire accidents - Automatic safety actions without human intervention - Real-time alerts and remote monitoring - Increased safety and peace of mind - Reduced risk of property damage and injury																		

	d. Cost advantage when compared to available solutions Uses low-cost, easily available components (Pico, ESP8266, basic sensors) More affordable than high-end smart safety systems Reduces long-term costs by preventing accidents and damage Modular design allows upgrades without replacing the full system							
26.	What is Technology readiness level of your invention? (Tick the appropriate TRL)							
	Research			Development			Deployment	
	TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8
	Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab 	Technology validated in a relevant environment (Industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (Industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified
27.	Any additional notes or remarks.							
	The invention emphasizes fail-safe, real-time safety control with minimal dependence on network connectivity. Its modular and low-cost design makes it suitable for large-scale adoption. The system can be further enhanced with additional sensors, backup power, and intelligent analytics, supporting future expansion and commercialization.							